IPv4 Addressing and Connectivity Lab Report

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Platform: Oracle VirtualBox
Systems: Ubuntu Server, Windows Server 2022
Focus: IPv4 addressing, subnet masks, default gateways, DNS, and connectivity testing

1. Objective

This lab builds a small virtual network to practise core IPv4 concepts and the first steps of network troubleshooting used in SOC and system administration work. The goals were to:

- Assign and verify IPv4 addresses and subnet masks on Linux and Windows.
- Understand the role of the default gateway and which interface carries Internet-bound traffic.
- Verify DNS resolution separately from raw IP connectivity.
- Test connectivity over three different virtual network types.
- Diagnose and fix a firewall-related connectivity failure.

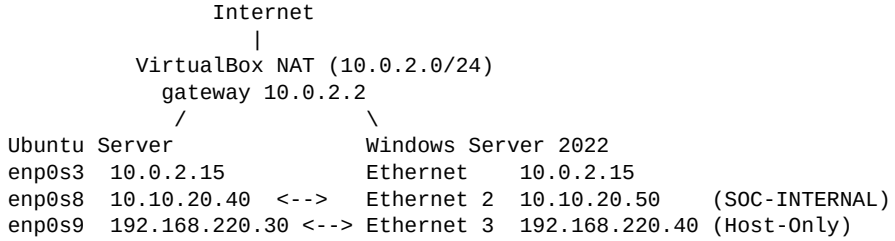
2. Lab Environment

Component	Details
Virtualization platform	Oracle VirtualBox
Linux system	Ubuntu Server (hostname `Ubuntu-Server-M009`)
Windows system	Windows Server 2022
Internal Network name	`SOC-INTERNAL`
Internal Network subnet	`10.10.20.0/24`
Host-Only subnet	`192.168.220.0/24`
VirtualBox NAT subnet	`10.0.2.0/24`

Each VM has three virtual network adapters:

Adapter type	Purpose
NAT	Internet access through VirtualBox's NAT service
Internal Network (`SOC-INTERNAL`)	Isolated VM-to-VM communication
Host-Only	Communication between the host computer and the VMs on a private virtual network

3. Network Design



4. IPv4 Addressing Plan

Machine	Interface	Purpose	IPv4 address	Subnet mask	Gateway
Ubuntu Server	`enp0s3`	NAT / Internet	`10.0.2.15`	`255.255.255.0` (/24)	`10.0.2.2`
Ubuntu Server	`enp0s8`	Internal Network	`10.10.20.40`	`255.255.255.0` (/24)	None
Ubuntu Server	`enp0s9`	Host-Only	`192.168.220.30`	`255.255.255.0` (/24)	None
Windows Server 2022	Ethernet	NAT / Internet	`10.0.2.15`	`255.255.255.0` (/24)	`10.0.2.2`
Windows Server 2022	Ethernet 2	Internal Network	`10.10.20.50`	`255.255.255.0` (/24)	None
Windows Server 2022	Ethernet 3	Host-Only	`192.168.220.40`	`255.255.255.0` (/24)	None

Duplicate NAT address. Both VMs show `10.0.2.15`. This is expected with VirtualBox's per-VM NAT mode: each VM sits in its own isolated NAT environment, so the duplicate does not cause an IP conflict. The Internal Network and Host-Only addresses are unique and are used for direct VM-to-VM traffic.

Why only the NAT adapter has a gateway. The Internal and Host-Only networks are used for local lab traffic only, so no gateway is needed on them. Configuring a gateway on more than one interface can create competing default routes, so the NAT adapter alone supplies the default route for Internet-bound traffic.

5. Key Concepts Applied

- **IPv4 address and subnet mask:** A /24 mask (`255.255.255.0`) means the first three octets identify the network and the last octet identifies the host, giving 254 usable addresses per subnet.
- **Default gateway:** The router a host uses for destinations outside its local subnet. Here it is `10.0.2.2`, VirtualBox's NAT gateway.
- **DNS:** Translates hostnames to IP addresses. Testing by IP (`8.8.8.8`) and by name (`google.com`) separates routing problems from DNS problems.
- **Private addressing:** All lab addresses are private and non-routable on the public Internet. NAT lets the guests reach the Internet without exposing their private addresses.
- **ICMP:** The protocol behind `ping`, used here as a basic reachability test.

6. Procedure and Commands

6.1 Ubuntu Server

Purpose	Command
Show interface addresses	`ip -br a`
Show routing table and default gateway	`ip route`
Test Internet by IP	`ping -c 4 8.8.8.8`
Test DNS and Internet	`ping -c 4 google.com`
Test Windows over Internal Network	`ping -c 4 10.10.20.50`
Test Windows over Host-Only	`ping -c 4 192.168.220.40`

6.2 Windows Server 2022

Purpose	Command
Show addressing, gateway, DHCP, DNS	`ipconfig /all`
Test Ubuntu over Internal Network	`ping 10.10.20.40`
Test Ubuntu over Host-Only	`ping 192.168.220.30`
Test Internet by IP	`ping 8.8.8.8`
Test DNS and Internet	`ping google.com`

7. Verification Evidence

7.1 Ubuntu interface addresses (`ip -br a`)

The command was run on `Ubuntu-Server-M009` and confirmed the planned configuration:

Interface	State	Address	Matches plan
`lo`	UNKNOWN	`127.0.0.1/8`	Loopback (expected)
`enp0s3`	UP	`10.0.2.15/24` (metric 100)	Yes, NAT
`enp0s8`	UP	`10.10.20.40/24`	Yes, Internal Network
`enp0s9`	UP	`192.168.220.30/24`	Yes, Host-Only

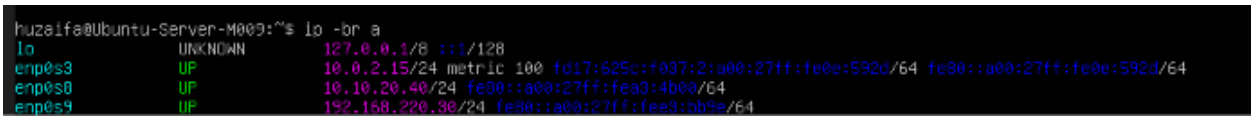


Figure 1: Output of ip -br a on Ubuntu Server.

All three adapters were up with the intended /24 addresses. Each interface also carries an IPv6 link-local address, which is not used in this lab.

7.2 Additional screenshots to attach

1. Ubuntu `ip route` showing the default route via `10.0.2.2` on `enp0s3`.
2. Ubuntu successful pings to `10.10.20.50`, `192.168.220.40`, and `google.com`.
3. Windows `ipconfig /all` showing all three adapters, the gateway, and the DNS server.
4. Windows successful pings to `10.10.20.40`, `192.168.220.30`, and `google.com`.
5. Optional: VirtualBox adapter settings showing NAT, `SOC-INTERNAL`, and Host-Only.

8. Connectivity Test Results

Source	Destination	Result	Interpretation
Ubuntu Server	`10.10.20.50`	6 packets received, 0% loss	Internal Network communication succeeded
Ubuntu Server	`192.168.220.40`	4 packets received, 0% loss	Host-Only communication succeeded
Ubuntu Server	`google.com`	4 packets received, 0% loss	DNS and Internet connectivity succeeded
Windows Server 2022	`10.10.20.40`	4 packets received, 0% loss	Internal Network communication succeeded

Source	Destination	Result	Interpretation
Windows Server 2022	`192.168.220.30`	4 packets received, 0% loss	Host-Only communication succeeded
Windows Server 2022	`google.com`	4 packets received, 0% loss (final test)	DNS and Internet connectivity succeeded

Some early ping attempts showed packet loss or no replies. Results above are from the tests run after troubleshooting.

9. Default Gateway and DNS

The Windows NAT adapter reported:

Setting	Value
IPv4 address	`10.0.2.15`
Subnet mask	`255.255.255.0`
Default gateway	`10.0.2.2`
DNS server	`192.168.1.1`
DHCP	Enabled

The Ubuntu routing table likewise showed its default route through `10.0.2.2` on `enp0s3`. The successful hostname pings on both systems confirmed that DNS resolution and Internet routing worked at the time of testing.

10. Troubleshooting

Symptom: Pings from Ubuntu to Windows Server received no replies.

Likely cause: Windows Firewall blocks inbound ICMP echo requests by default.

Resolution:

1. Added an inbound ICMPv4 Echo Request rule on Windows Server, scoped to the lab subnets `10.10.20.0/24` and `192.168.220.0/24`.
2. Re-tested Ubuntu to Windows on both the Internal and Host-Only networks: successful.
3. Tested Windows to Ubuntu on both networks: successful.
4. Re-tested Internet and DNS on both systems: successful after earlier intermittent packet loss.

Approach: Check addressing first, then routing, then DNS, then firewall rules, working from the local network outward.

11. Security Considerations

- ICMP was allowed only to support diagnostics. The rule was limited to the lab subnets rather than opened to any source.
- On a production server, ICMP rules should be restricted to trusted networks and enabled only when needed.
- A successful ping proves ICMP reachability only. It does not prove that any application service or port is reachable or secure.

- Host-Only and Internal networks keep lab traffic isolated from the physical network, which limits exposure while testing.
- Public IP addresses observed during testing are omitted from this report. Public addresses change with the provider and are best kept out of published documentation.

12. Conclusion

The lab produced a working three-network environment in which Ubuntu Server and Windows Server 2022 communicated in both directions over the Internal and Host-Only networks and reached the Internet with working DNS. It reinforced the habit of checking interface addresses, routes, DNS settings, and firewall rules in order when diagnosing connectivity problems.

13. Skills Demonstrated

- IPv4 addressing and /24 subnetting
- Multi-adapter VM networking in VirtualBox (NAT, Internal, Host-Only)
- Linux networking tools: `ip`, `ping`
- Windows networking tools: `ipconfig`, `ping`, Windows Firewall rules
- Structured network troubleshooting and documentation

Appendix: Quick Command Reference

Task	Linux	Windows
View addresses	`ip -br a`	`ipconfig /all`
View routes / gateway	`ip route`	`route print`
Test reachability	`ping -c 4 <target>`	`ping <target>`
Test DNS	`ping -c 4 google.com`	`ping google.com`